

Fascia-focused Pilates training

Elizabeth Larkam

Note Elizabeth Larkam has produced a compilation of short videos to help you gain a deeper understanding. Watch these videos free of charge at www.pilatesanytime.com/fascia. To redeem your free access use code **FASCIA**. These short videos were compiled as a companion to her book *Fascia in Motion: Fascia-focused Movement for Pilates* (Handspring, 2017).

Introduction

The movement program created by Joseph Hubertus Pilates (1880–1967) served the bedridden after World War I broke out in 1914, when he taught in a camp for enemy aliens in Lancaster, England. J. H. Pilates took springs from beds, creating his earliest rehabilitation equipment for the internees (Pilates Method Alliance, 2013).

Less than a century later, it was reported that Pilates was the fastest growing activity in the country, with 8.6 million Americans participating in 2009 (Sporting Goods Manufacturers Association, 2010). Participation increased by 456% between 2000 and 2009 (Rovell, 2010). Countless millions have engaged in the practice of Pilates worldwide. In 2015, Statista reported that the number of United States participants aged 6 years and older in Pilates training amounted to ~8.6 million (Statista, 2015).

Pilates programs, informed by principles of fascia-oriented training (Müller & Schleip, 2011) (Chapter 24) have been shown to improve standing balance, seated balance and proximal control of gait for polytrauma members of the military returning from Iraq and Afghanistan with amputations (Moore, 2009), trauma, traumatic stress, brain injury and vestibular dysfunction (Larkam, 2013). Soldiers from Denmark, Israel and the United States demonstrate their fascia-focused Pilates programs, informed by the

organization of the myofascial continuities (Myers, 2020) (Chapter 36) (Pilates Method Alliance Video, 2013).

History of the Pilates method

The Pilates method has been in continuous use in the United States since 1926 when Joseph Pilates and his wife Clara immigrated to New York City from Germany. In 2010 there were 8653000 Pilates participants in the United States (Sporting Goods Manufacturers Association, 2010) and countless millions engaged in the practice of Pilates worldwide. For more than 40 years, J. H. Pilates created and documented a comprehensive movement system that reflected his own physical training and the physical culture of the times. In his youth, he practiced various physical regimens to overcome childhood ailments of rickets, asthma and rheumatic fever. The movement system Joseph Pilates created was influenced by his practice of bodybuilding, gymnastics, skiing, diving, martial arts and boxing. Joseph Pilates did not have any clinical or medical credentials (PMA, 2013). He was self-taught through movement experience, observation and reading. His ideas regarding spinal alignment (Pilates, 1934) were markedly different from the generally accepted view that optimal spine organization requires a cervical and lumbar lordosis and a thoracic kyphosis. In fact, Joseph Pilates designed his entire movement system to straighten spinal curves, reflecting his belief that

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“the normal spine should be straight to successfully function according to the laws of nature in general and the law of gravity, in particular.... Proper carriage of the spine is the only natural preventive against abdominal obesity, shortness of breath,

asthma, high and low blood pressure and various forms of heart disease. It is safe to say that none of the ailments here enumerated can be cured until the curvatures of the spine have been corrected” (Pilates, 1934).

Table 39.1 Publications that inform fascia-focused Pilates movement

Publications (alphabetically by last name)	Branch of knowledge
I: Models of movement	
Beach, P. (2010) <i>Muscles and Meridians: the manipulation of shape</i> . Edinburgh, UK: Churchill Livingstone.	Osteopathy, Traditional Chinese medicine
Martin, D.-C. (2016) <i>Living biotensegrity: interplay of tension and compression in the body</i> . Munich, Germany: Keiner.	Physics, Chinese movement arts
Scarr, G. (2018) <i>Biotensegrity: the structural basis of life</i> , 2nd edn. Edinburgh, UK: Handspring. Stephen Levin introduced the term biotensegrity in 1975.	Osteopathy
II: Architecture and anatomy of fascia	
Guimberteau, J.-C. & Armstrong, C. (2015) <i>Architecture of human living fascia: the extracellular matrix and cells revealed through endoscopy</i> . Edinburgh, UK: Handspring.	General medicine, surgery
Stecco, C. (2015) <i>Functional atlas of the human fascial system</i> . Edinburgh, UK: Churchill Livingstone.	General medicine, orthopedic surgery, human anatomy
Stecco, A., et al. (2016) <i>Fascial disorders: implications for treatment</i> . <i>American Academy of Physical Medicine & Rehabilitation Journal</i> 8(2): 161–168.	General medicine with specialty in physiatry
Van der Wal, J. (2009) The architecture of the connective tissue in the musculoskeletal system—an often overlooked functional parameter as to proprioception in the locomotor apparatus. <i>International Journal of Therapeutic Massage & Bodywork</i> 2(4): 9–23.	Fascia research in medical laboratory, general medicine
III: Fascia-informed manual therapies	
Chaitow, L. (2018) <i>Fascial dysfunction: manual therapy approaches</i> . Edinburgh, UK: Handspring.	Osteopathy
Myers, T. W. (2020) <i>Anatomy trains®: myofascial meridians for manual & movement therapists</i> (3rd ed.). Edinburgh, UK: Elsevier.	Structural integration
Schwind, P. (2006) <i>Fascial and membrane technique</i> . Edinburgh, UK: Churchill Livingstone. Schwind studied with Jean-Pierre Barral.	Rolfing®
Stecco, L. & Stecco, C. (2014) <i>Fascial manipulation® for internal dysfunctions</i> . Stuttgart, Germany: Schattauer GmbH.	Physiotherapy, general medicine, human anatomy
IV: Fascia-informed theory and movement applications	
Avison, J. S. (2015) <i>Yoga: fascia anatomy and movement</i> . Edinburgh, UK: Handspring.	Yoga Structural integration
Schleip, R. (2015) <i>Fascia in sport and movement</i> . Edinburgh, UK: Handspring. Robert Schleip and Thomas W. Myers coined the term “Fascial Fitness®”	Fascia research, Rolfing®

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Table 39.2 Pilates movement principles correlate with fascia-focused training criteria

The eight movement principles of the Pilates method (Lessen, 2013) correlate substantially with suggested fascia-focused training criteria (Larkam, 2017)

Pilates principles	Fascia-focused movement criteria
Whole-body movement	Whole-body movements simultaneously involving large areas of the neuromyofascial system Whole-body continuity — connect trunk to limbs and limbs to trunk — connect deep structures to superficial ones and superficial structures to deep ones
Centering	Movement initiation — connect proximal structures to distal ones and distal structures to proximal ones
Precision	Kinesthetic acuity — kinesthesia (dynamic proprioception), the aptitude to sense the position and movement of the limbs and trunk
Control	Proprioceptive refinement — proprioceptive sensations are connected with position, tendon and muscle sensations
Concentration	Interoception — interstitial nerves in fascia serve an interoceptive, rather than proprioceptive or nociceptive function. Stimulation of those free nerve endings provides information about the condition of the body in search for homeostasis in relation to physiological needs. Interoceptive signaling is associated with feelings like warmth, nausea, hunger, soreness, effort, heaviness or lightness. Perceptions about internal somatic sensations are associated with emotional preferences and feelings (Schleip, 2015)
Breathing	Flowing movement sequences
Balanced muscle development (optimal activation of the neuromyofascial system)	Multidirectional movements with slight changes of angle
Rhythm	Preparatory countermovement Dynamic stretching — slow and fast tempo variations
These fascia-focused movement criteria do not correlate with specific Pilates principles. However, fascia-focused Pilates movement may involve them	
	Develop glide within the three-dimensional neuromyofascial system Force/load transfer — movements that distribute force/load through the neuromyofascial system Facilitate tissue hydration Develop regular lattice with crimp Develop elastic recoil Stimulate tissue renewal Develop tissue resilience Movements are sustainable for motor control refinement and collagen remodeling Access all appropriate myofascial continuities in optimal sequence for client profile Movements encourage awareness of and embodiment of applied biotensegrity

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Selected mat and reformer exercises provide examples of the Pilates principles

This chapter examines the effectiveness and the limitations of the original Pilates method to fulfill the principles of fascia-focused training.

This photograph suggests that the original Pilates method of unique movement sequences and equipment created a foundation from which contemporary expressions of Pilates can be customized to train fascia. The chapter includes one example of how the J. H. Pilates mat exercise Leg Pull Front provides a foundation for the contemporary Pilates Reformer exercise, Jumping

Table 39.3 Selected mat and Reformer exercises provide examples of the Pilates principles		
Pilates principles	Pilates mat exercise	Reformer exercise
Rhythm	The One-Leg Stretch with Knees Extended (Pilates Anytime Mat Workout (2016) #2507)	Elastic recoil training with jumpboard
Breathing	The Roll Up (Figure 39.2A2)	Standing side splits
Rhythm	The Saw (Figure 39.2E2)	Knee stretch series Standing/knees off
Rhythm	Fast: The One Leg Kick (Figure 39.2C2) Slow: The Roll Over With Legs Spread (Both Ways) (Figure 39.2B1)	Fast: Stomach Massage Slow: Side Stretch
Whole body movement	Side Bend Twist (Figure 39.2F3)	Long-Stretch
Balanced muscle development (optimal activation of the neuromyofascial system)	The Cork-Screw (Figure 39.2E1)	Feet in straps
Centering	The Teaser (Figure 39.2A6)	Short box advanced abdominals
Concentration control	The Leg-Pull (Figure 39.2F2)	Control back (face ceiling)
Precision	The Hip Twist With Stretched Arms (Figure 39.2E4)	Front splits – hands up
These fascia-oriented training principles do not correlate with specific Pilates principles; however, fascia-focused Pilates movement may fulfill these benefits	The Spine Stretch (Figure 39.2A5)	Spine-Massage
	The Spine Twist (Figure 39.2E3)	Cleopatra
	Mat practice 20 min twice per week for between 7 months and 2 years.	Reformer practice 20 min twice per week for between 7 months and 2 years
“If you will faithfully perform your Contrology exercises regularly only four times a week for just three months as outlined in Return to Life, you will find your body development approaching the ideal, accompanied by renewed mental vigor and spiritual enhancement.” Joseph H. Pilates (1945–2010).		

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Figure 39.1

Joseph H. Pilates instructing a client in Hanging Pull Up on his trapeze table at Joe's Gymnasium, the 8th Avenue Studio, New York, NY, October 1961. The Universal Reformer is visible in the background.

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in Quadruped, which fulfills the principles of fascia-focused training. The Pilates movement principles are explained in *The Pilates Method Alliance Pilates Certification Exam Study Guide* (PMA, 2013), which states: “The body is organised to move by **centering**. **Balanced muscle development** allows efficient movement and proper joint mechanics. Constant mental **concentration** is required to fully develop the body. **Precision**, meaning exact, defined, specific, intentional movement is necessary for correct form. Only a few repetitions of each exercise are appropriate so that each repetition can be performed with the greatest **control**, using only the necessary muscles and effort necessary for each movement. **Breathing** promotes natural movement and **rhythm** and stimulates muscles to greater activity. Performance of the Pilates exercises is distinguished by always using the **whole body**.”

Mat exercises provide the foundation of the Pilates method

Thirty-four mat exercises form the foundation of the movement system that Joseph Pilates called

“Contrology”. Joseph Pilates states his philosophy in his 1945 book, *Return to Life*: “Faithfully perform your Contrology exercises only four times a week for just three months...you will find your body development approaching the ideal, accompanied by renewed mental vigor and spiritual enhancement” (Pilates, 1945). Careful study of the mat exercises indicates that, although specific Pilates mat exercises may satisfy the fascia-focused training criteria “Multidirectional Movements with slight changes in angle” (Table 39.2), the majority of exercises will not, given their single plane orientation. Of the mat exercises pictured in *Return to Life Through Contrology*, 15 are organized in flexion in the sagittal plane. Only eight exercises emphasize extension in the sagittal plane. Lateral flexion of the spine is represented by two exercises. Three exercises combine flexion and rotation. Spinal rotation is represented by The Spine Twist. Joseph Pilates intended the remaining five exercises to be performed in a posterior pelvic tilt and lumbar flexion in order to flatten the lumbar lordosis. However, they may be practiced in a neutral pelvis and a neutral lumbar spine.

Given that 29 of the mat exercises are open kinematic chain for the lower extremity, these mat exercises are not optimal for developing elastic recoil in full weight bearing. Only four exercises are closed kinematic chain for both upper and lower extremities. Joseph Pilates performed many of the mat exercises with rather vigorous elastic bounces in the end ranges of available motion (Pilates, 1932–1945) so even his open kinematic chain exercises can be used to develop elastic recoil (Chapter 8).

Joseph Pilates was a prolific inventor of exercise apparatus

Throughout his 4 decades of work in Joe's Gymnasium, Joseph Pilates continued to develop

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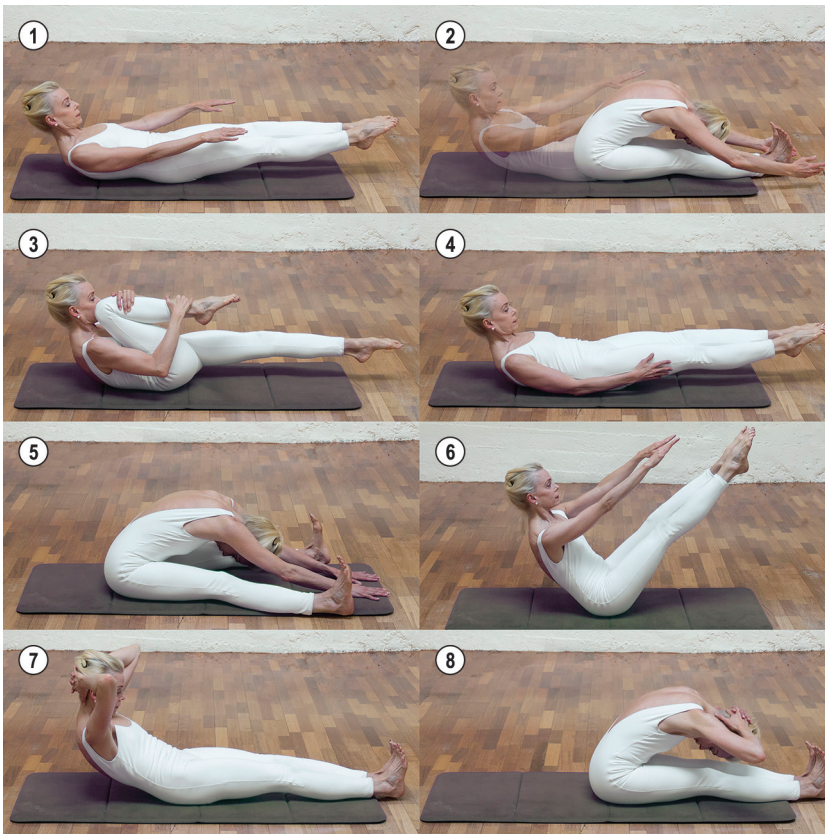


Figure 39.2 A

Seven Pilates mat exercises demonstrate spine flexion in supine and sitting. These movements emphasize activation of the Deep Front (Figure 36.7) and Superficial Front (Figure 36.2) myofascial continuities. From top left to bottom right: (1) The Hundred, (2) The Roll-Up, (3) The One Leg Stretch, (4) The Double Leg Stretch, (5) The Spine Stretch, (6) The Teaser, (7, 8) The Neck Pull.

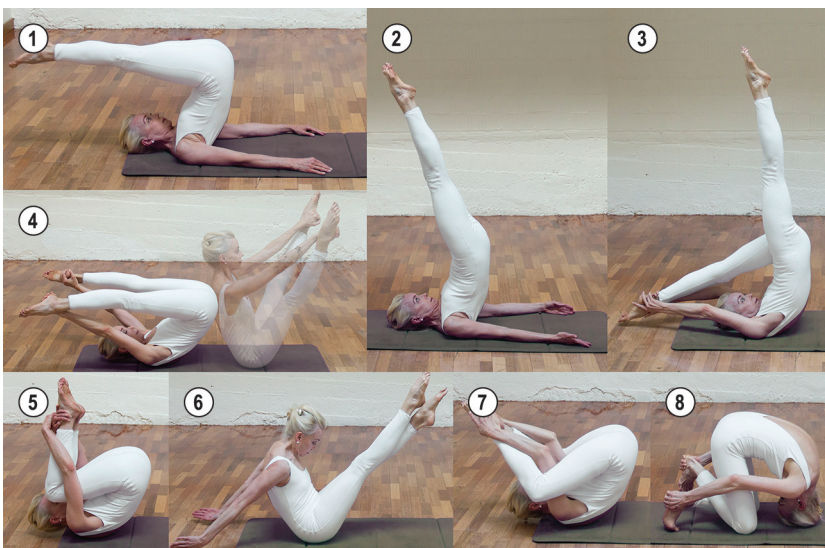


Figure 39.2 B

Eight Pilates mat exercises demonstrate spine flexion with weight bearing on the upper thoracic area. These movements emphasize synergistic activation of the Superficial Front (Figure 36.2) and Superficial Back (Figure 36.1) myofascial continuities. From top left to bottom right: (1) The Roll Over, (2) Rocker with Open Legs, (3) The Jack-Knife, (4) The Control Balance, (5) Rolling Back, (6) The Boomerang, (7) The Seal, (8) The Crab.

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Figure 39.2 C

Eight Pilates mat exercises demonstrate spine extension. These movements emphasize synergistic activation of the Superficial Back (Figure 36.1) myofascial continuity in conjunction with the Deep Front (Figure 36.7) myofascial continuity. From top left to bottom right: (1) The Swan-Dive, (2) The One Leg Kick, (3) The Double Kick, (4) The Scissors, (5) The Bicycle, (6) The Shoulder Bridge, (7) Swimming, (8) The Rocking.



Figure 39.2 D

Two Pilates mat exercises demonstrate lateral flexion of the spine. These movements emphasize activation of the Lateral myofascial continuity (Figure 36.3). Left to right: (1, 2) The Side Kick, (3) The Side Kick Kneeling.

(Continued)

his movement program, inventing over 14 exercise apparatus that provide assistance, resistance and complexity to the organization required by the mat exercises. The equipment frames provide a number of places to which springs or cords may be attached, creating movement environments that support a variety of vectors, making it possible to satisfy all the criteria of fascia-focused training listed in Table 39.2. Joseph Pilates created a unique movement repertoire for each of his inventions.

Contemporary evolution of the Pilates method serves client and patient diversity

A century has passed since Joseph Pilates began working in a New York City boxer's training gym, teaching a diverse clientele. The proliferation of applications of Pilates techniques has resulted in an expansion of protocols for a

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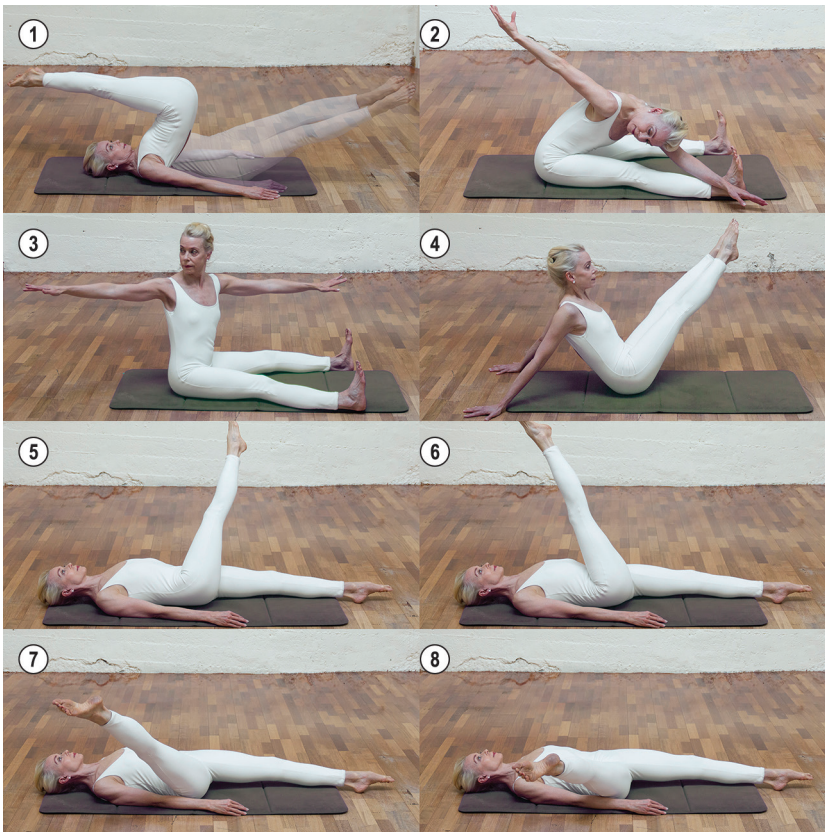


Figure 39.2 E

Five Pilates mat exercises demonstrate spine rotation. These movements emphasize activation of the Spiral myofascial continuity (Figure 36.4). Top left to bottom right: **(1)** The Cork-Screw, **(2)** The Saw, **(3)** The Spine Twist, **(4)** The Hip Twist with Stretched Arms, **(5–8)** The One Leg Circle (Both Ways).

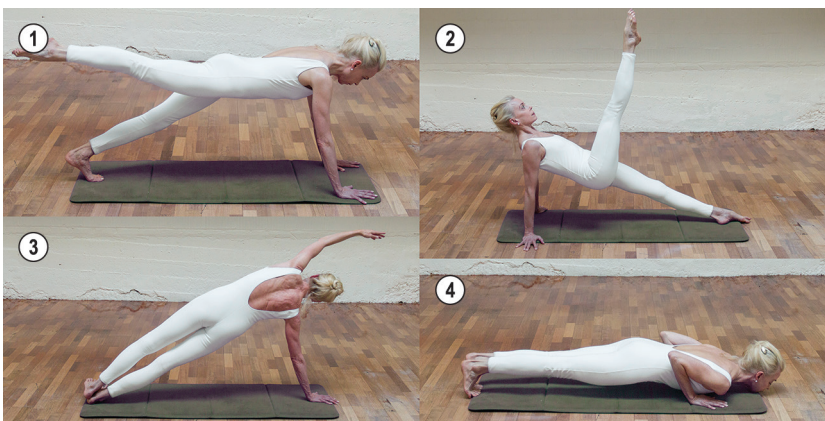


Figure 39.2 F

Four Pilates mat exercises demonstrate closed kinematic chain contact of the hands and feet with the mat. These movements activate the entire neuromyofascial system (Figure 36.1–36.7). From top left to bottom right: **(1)** The Leg-Pull – Front, **(2)** The Leg-Pull, **(3)** The Side Bend, **(4)** The Push Up.

wide range of diagnoses, conditions and performance goals. Research on the properties and function of the neuromyofascial system, as

well as the creation of Fascial Fitness (Chapter 24), influences interest in how to train fascia in Pilates.

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Table 39.4 The myofascial continuities^a and J. H. Pilates mat

Use this Pilates mat fascia-focused training guide to plan your mat classes and individual practice for thorough fascial training. Each exercise cultivates at least one of the fascia-focused movement criteria and activates one primary and several secondary myofascial continuities

Fascia-focused movement criteria		The myofascial continuities ^a						
Movement requirements		Biotensegrity model						
1. Movements are sustainable for motor control refinement and collagen remodeling		b "A tensegrity structure provides a global response to a local mechanical stress. The result is a degree of independence from the force of gravity. Without a tensegrity model, our fibrillar structure would collapse under gravitational force. With the tensegrity model, our structure absorbs and disperses the compression by spreading the load throughout the entire network, including the structures at the periphery" (Guimberteau and Armstrong, 2015).						
2. Movements encourage awareness of and embodiment of the biotensegrity model ^b		"The value of the Tensegrity model is <i>not</i> that it necessarily changes a particular approach to treatment but that it provides a <i>better means to visualize the mechanics of the body in the light of new understandings about functional anatomy</i> " (Scarr, 2018).						
3. Access all appropriate myofascial continuities in optimal sequence for client profile		Superficial Back	Functional	Lateral	Spiral	Deep Front	Arm	Superficial Front
Movement intent		The Swan-Dive (Figure 39.2C1)	Rocker With Open Legs (Figure 39.2B2)	The Side Bend (Figure 39.2F3)	The Spine Twist (Figure 39.2E3)	Rolling Back (Figure 39.2B5)	The Leg-Pull (Figure 39.2F2)	The Bicycle (Figure 39.2C5)
4. Interoception — Intersitil nerves in fascia serve an interoceptive, rather than proprioceptive or nociceptive function. Stimulation of those free nerve endings provides information about the condition of the body in search for homeostasis in relation to physiological needs. Interoceptive signaling is associated with feelings like warmth, nausea, hunger, soreness, effort, heaviness or lightness. Perceptions about internal somatic sensations are associated with emotional preferences and feelings (Schleip, 2015).								
5. Proprioceptive refinement — Proprioceptive sensations are connected with position, tendon and muscle sensations		The Leg-Pull (Figure 39.2F2)	The Jack-Knife (Figure 39.2B3)	Rocker With Open Legs (Figure 39.2B2)	The Cork-Screw (Figure 39.2E1)	The Teaser (Figure 39.2A6)	The Leg-Pull – Front (Figure 39.2F1)	The Roll Up (Figure 39.2A2)
6. Kinesthetic acuity — Kinesthesia (dynamic proprioception), the aptitude to sense the position and movement of the limbs and trunk		Swimming (Figure 39.2C7)	Rolling Back (Figure 39.2B5)	The Side Kick Kneeling (Figure 39.2D3)	The Hip Twist With Stretched Arms (Figure 39.2E4)	The Double Leg Stretch (Figure 39.2A4)	The Side Bend (Figure 39.2F3)	The One Leg Stretch (Figure 39.2A3)

(Continued)

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Table 39.4 (Continued) The myofascial continuities^a and J. H. Pilates mat

Use this Pilates mat fascia-focused training guide to plan your mat classes and individual practice for thorough fascial training. Each exercise cultivates at least one of the fascia-focused movement criteria and activates one primary and several secondary myofascial continuities

Fascia-focused movement criteria								
Movement attributes								
7. Whole-body movements simultaneously involving large areas of the neuromyofascial system	The Shoulder Bridge (Figure 39.2C6)	The One Leg Kick (Figure 39.2C2)	The Side Bend (Figure 39.2F3)	The Cork-Screw (Figure 39.2E1)	The Neck Pull (Figure 39.2A8)	The Push Up (Figure 39.2F4)	The Crab (Figure 39.2B8)	
	Swimming (Figure 39.2C7)	The Leg-Pull (Figure 39.2F1)	The Side Kick (Figure 39.2D1, 2)	The Hip Twist With Stretched Arms (Figure 39.2E4)	The Scissors (Figure 39.2C4)	The Side Bend (Figure 39.2F3)	The Double Leg Stretch (Figure 39.2A4)	
8. Whole-body continuity								
— Connect trunk to limbs and limbs to trunk								
— Connect deep structures to superficial ones and superficial structures to deep ones								
9. Movement initiation								
— Connect proximal structures to distal ones and distal structures to proximal ones	The Rocking (Figure 39.2C8)	Swimming (Figure 39.2C7)	The Side Kick (Figure 39.2D1, 2)	The Cork-Screw (Figure 39.2E1)	Rocker With Open Legs (Figure 39.2B2)	The Boomerang (Figure 39.2B6)	The Seal (Figure 39.2B7)	
10. Flowing movement sequences								
	The Swan-Dive (Figure 39.2C1)	The Leg-Pull – Front (Figure 39.2F1)	The Side Kick Kneeling (Figure 39.2D3)	The Saw (Figure 39.2E2)	The Jack-Knife (Figure 39.2B3)	Swimming (Figure 39.2C7)	Rolling Back (Figure 39.2B5)	
11. Multidirectional movements with slight changes of angle								
	The Bicycle (Figure 39.2C5)	The One Leg Circle (Both Ways) (Figure 39.2E5-8)	The Hip Twist With Stretched Arms (Figure 39.2E4)	The Cork-Screw (Figure 39.2E1)	The Hip Twist With Stretched Arms (Figure 39.2E4)	The Hip Twist With Stretched Arms (Figure 39.2E4)	The One Leg Circle (Both Ways) (Figure 39.2E5-8)	
12. Force/load transfer								
— Movements that distribute force/load through the neuromyofascial system	The Rocking (Figure 39.2C8)	The Rocking (Figure 39.2C8)	The Side Kick (Figure 39.2D1, 2)	The Hip Twist With Stretched Arms (Figure 39.2E4)	The Seal (Figure 39.2B7)	The Side Bend (Figure 39.2F3)	The Seal (Figure 39.2B7)	
	The Swan-Dive (Figure 39.2C1)	The Leg-Pull – Front (Figure 39.2F1)	Rocker With Open Legs (Figure 39.2B2)	The Cork-Screw (Figure 39.2E1)	The Push Up (Figure 39.2F4)	The Push Up (Figure 39.2F4)	The Crab (Figure 3.3H)	

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Table 39.4 (Continued) The myofascial continuities^a and J. H. Pilates mat

Use this Pilates mat fascia-focused training guide to plan your mat classes and individual practice for thorough fascial training. Each exercise cultivates at least one of the fascial-focused movement criteria and activates one primary and several secondary myofascial continuities

Fascia-focused movement criteria							
13. Preparatory counter movement	The Double Kick (Figure 39.2C3)	The Side Kick Kneeling (Figure 39.2D3)	The Side Kick (Figure 39.2D1,2)	The Spine Twist (Figure 39.2E3)	The Shoulder Bridge (Figure 39.2C6)	The Spine Twist (Figure 39.2E3)	The Bicycle (Figure 39.2C5)
14. Facilitate tissue hydration	Approximately two-thirds of the volume of fascial tissues is made up of water. During application of mechanical load, whether by stretching or via local compression, a significant amount of water is pushed out of the more stressed zones. Application of external loading of fascial tissues can result in a refreshed hydration with improvement in the viscoelastic tissue properties (Schleip and Baker, 2015, p. 8).						
15. Develop glide within the three-dimensional neuromyofascial system	The normal glide of the collagen layers over one another may be hampered by loose connective tissue that has accumulated a variety of waste products. Failure of loose connective tissue gliding occurs in cases of overuse syndrome, trauma and surgery. Increase of local temperature contributes to fascial gliding. When the temperature is increased to over 40°C, the three-dimensional superstructure of hyaluronan chains break down progressively. This decreases hyaluronan viscosity of the loose connective tissue that is present within and beneath deep fasciae and muscles. The temporary increase in temperature can be provided by massage, movement or a warm shower. It is possible that during sleep, immobilization may cause increased hyaluronan viscosity and consequently a stiffness of the deep fasciae and muscles (Stecco, 2015, p. 60, p. 64).						
16. Develop regular lattice with crimp	The microstructure of collagen fibers shows undulations called crimp, which is reminiscent of elastic springs. When people age or when fascial fibers are immobilized, the structure of the fiber appears to loose crimp and springiness. Regular proper exercise loading of the fascial fibers may induce a more youthful collagen architecture with wavy fibers and increased elastic storage capacity (Schleip and Baker, 2015, p. 7).						
17. Develop elastic recoil							
18. Stimulate tissue renewal	The renewal speed of the body-wide fascial network is quite slow, with a half-life between months and years rather than days or weeks. During the first 3 hours after appropriate exercise loading, collagen synthesis is increased. However, collagen degradation is also increased. During the first one and a half days following exercise loading, collagen degradation outweighs collagen synthesis. Forty-eight hours after exercise, net synthesis of collagen production becomes positive (Schleip and Baker, 2015, pp. 8–9). The mobile, adaptable, fibrillar network with its intersecting fibers develops a mechanical harmony that is lost when healthy tissue is damaged. The body's repair mechanisms are unable to restore the fibrillar network in the damaged area to its original condition. The replacement tissue is of poor quality but can be improved by early mobilization of the injured area by manual therapy to enhance the flexibility of the scar tissue (Guimberteau and Armstrong, 2015, p. 162).						
19. Develop tissue resilience	According to the bidensegrity model, the neuromyofascial system accommodates movement as a self-adjusting whole. The myofascial meridians map out a set of connective tissues that form the outer tensional network that pulls in on the skeleton. The "isolated muscle theory" that has predominated our thinking has limited our perception of this body-wide "give" that is essential to resilience (Myers, p. 46; Ch. 6 in Schleip and Baker, 2015).						

^a *Anatomy Trains* by Thomas W. Myers (2015) introduces the term myofascial meridians. In this chapter I replace "meridians" with "continuities".

^a *Anatomy Trains* by Thomas W. Myers (2015) introduces the term myofascial meridians. In this chapter I replace 'meridians' with 'continuities'.

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Table 39.5 The myofascial continuities^a and J. H. Pilates Reformer

Use this Pilates Reformer fascia-focused training guide to plan your Reformer classes and individual practice for thorough fascial training. Each exercise cultivates at least one of the fascia-focused movement criteria and activates one primary and several secondary myofascial continuities

Fascia-focused movement criteria															
Movement requirements	Biotensegrity model														
1. Movements are sustainable for motor control refinement and collagen remodeling	^a "A tensegrity structure provides a global response to a local mechanical stress. The result is a degree of independence from the force of gravity. Without a tensegrity model, our fibrillar structure would collapse under gravitational force. With the tensegrity model, our structure absorbs and disperses the compression by spreading the load throughout the entire network, including the structures at the periphery" (Guimberteau and Armstrong, 2015).														
2. Movements encourage awareness of and embodiment of the biotensegrity model ^b	"The value of the Tensegrity model is not that it necessarily changes a particular approach to treatment but that it provides a better means to visualize the mechanics of the body in the light of new understandings about functional anatomy" (Scarr, 2014).														
	The myofascial continuities^a														
3. Access all appropriate myofascial continuities in optimal sequence for client profile	<table><tr><th>Superficial Back</th><th>Functional</th><th>Lateral</th><th>Spiral</th><th>Deep Front</th><th>Arm</th><th>Superficial Front</th></tr><tr><td>Thigh-Stretch</td><td>Balance-Control 3</td><td>Swimming – Backstroke</td><td>Horseback 2</td><td>Control Stretch</td><td>Long-Stretch</td><td>Balance-Control 4</td></tr></table>	Superficial Back	Functional	Lateral	Spiral	Deep Front	Arm	Superficial Front	Thigh-Stretch	Balance-Control 3	Swimming – Backstroke	Horseback 2	Control Stretch	Long-Stretch	Balance-Control 4
Superficial Back	Functional	Lateral	Spiral	Deep Front	Arm	Superficial Front									
Thigh-Stretch	Balance-Control 3	Swimming – Backstroke	Horseback 2	Control Stretch	Long-Stretch	Balance-Control 4									
Movement intent															
4. Interception — Intersititial nerves in fascia serve an interoceptive, rather than proprioceptive or nociceptive, function. Stimulation of those free nerve endings provides information about the condition of the body in search for homeostasis in relation to physiological needs. Interoceptive signaling is associated with feelings like warmth, hunger, soreness, effort, heaviness or lightness. Perceptions about internal somatic sensations are associated with emotional preferences and feelings (Schleip, 2015).															
5. Proprioceptive refinement — Proprioceptive sensations are connected with position, tendon and muscle sensations	<table><tr><th>Breaststroke</th><th>Horseback 1</th><th>Control Stretch – Lifting</th><th>Horseback 1</th><th>Long-Stretch</th><th>Rowing – Back</th><th>Spine-Massage</th></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>	Breaststroke	Horseback 1	Control Stretch – Lifting	Horseback 1	Long-Stretch	Rowing – Back	Spine-Massage							
Breaststroke	Horseback 1	Control Stretch – Lifting	Horseback 1	Long-Stretch	Rowing – Back	Spine-Massage									
6. Kinesthetic acuity — Kinesthesia (dynamic proprioception), the aptitude to sense the position and movement of the limbs and trunk	<table><tr><th>Long-Backstretch</th><th>Back-Bending 3</th><th>Control Stretch – Bending</th><th>Chest-Expansion</th><th>Horseback 2 & 3</th><th>Rowing – Front</th><th>Semicircles Up & Down</th></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>	Long-Backstretch	Back-Bending 3	Control Stretch – Bending	Chest-Expansion	Horseback 2 & 3	Rowing – Front	Semicircles Up & Down							
Long-Backstretch	Back-Bending 3	Control Stretch – Bending	Chest-Expansion	Horseback 2 & 3	Rowing – Front	Semicircles Up & Down									

Fascia-focused Pilates training

Table 39.5 (Continued) The myofascial continuities^a and J. H. Pilates Reformer

Use this Pilates Reformer fascia-focused training guide to plan your Reformer classes and individual practice for thorough fascial training. Each exercise cultivates at least one of the fascia-focused movement criteria and activates one primary and several secondary myofascial continuities

Fascia-focused movement criteria									
Movement attributes									
7. Whole-body movements simultaneously involving large areas of the neuro-myofascial system	Swan-Dive	Russian-Stretch 2	Side Stretch & Control Stretch	Twist	Twist	Twist	Long-Backstretch	Russian-Stretch 1	
	Rocking	Russian-Stretch 1	Long Spine Stretch	Corkscrew	Back-Bending 2	Tendon-Stretch	Swimming – Backstroke		
8. Whole-body continuity — Connect trunk to limbs and limbs to trunk — Connect deep structures to superficial ones and superficial structures to deep ones	Rocking & Pull	Balance-Control 1	Side Stretch & Control Stretch	Chest-Expansion	Stomach Massage	Balance-Control 1	Rowing – Front		
	Back-Bending 1	Russian-Stretch 1 & 2	Swimming – Backstroke	Twist	Back-Bending 3	Triceps	Triceps		
9. Movement initiation — Connect proximal structures to distal ones and distal structures to proximal ones	Swimming	Corkscrew	Side Stretch & Control Stretch	Corkscrew	Corkscrew	Twist	Corkscrew		
	Lifting	Balance-Control 1	Long Spine Stretch	Horseback 3	Overhead	Side Stretch & Control Stretch	Arches (Footwork)		
10. Flowing movement sequences	Down-Stretch	Twist	Side Stretch & Control Stretch	Russian-Stretch 2	Tendon-Stretch	Balance-Control 3	Spine-Massage		
11. Multidirectional movements with slight changes of angle									
12. Force/load transfer — Movements that distribute force/load through the neuromyo-fascial system									

(Continued)

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Table 39.5 (Continued) The myofascial continuities^a and J. H. Pilates Reformer Use this Pilates Reformer fascia-focused training guide to plan your Reformer classes and individual practice for thorough fascial training. Each exercise cultivates at least one of the fascia-focused movement criteria and activates one primary and several secondary myofascial continuities						
Fascia-focused movement criteria						
13. Preparatory counter movement	Swan-Dive	Swimming	Control Stretch – Bending	Horseback 2	Feet & Knees (Footwork)	Swan-Dive Tiger-Stretch
14. Dynamic stretching — Slow and fast tempo variations	Down-Stretch	Side Stretch & Control Stretch	Side Stretch & Control Stretch	Corkscrew	Tiger-Stretch	Semicircles Up & Down
Movement outcomes						
15. Facilitate tissue hydration	Approximately two-thirds of the volume of fascial tissues is made up of water. During application of mechanical load, whether by stretching or via local compression, a significant amount of water is pushed out of the more stressed zones. Application of external loading of fascial tissues can result in a refreshed hydration with improvement in the viscoelastic tissue properties (Schleip and Baker, 2015, p. 8).					
16. Develop glide within the three-dimensional neuro-myofascial system	The normal glide of the collagen layers over one another may be hampered by loose connective tissue that has accumulated a variety of waste products. Failure of loose connective tissue gliding occurs in cases of overuse syndrome, trauma and surgery. Increase of local temperature contributes to fascial gliding. When the temperature is increased to over 40°C, the three-dimensional superstructure of hyaluronan chains break down progressively. This decreases hyaluronan viscosity of the loose connective tissue that is present within and beneath deep fascia and muscles. The temporary increase in temperature can be provided by massage, movement or a warm shower. It is possible that during sleep, immobilization may cause increased hyaluronan viscosity and consequently a stiffness of the deep fascia and muscles (Stecco, 2015, p. 60, p. 64).					
17. Develop regular lattice with crimp	The microstructure of collagen fibers shows undulations called crimp, which is reminiscent of elastic springs. When people age or when fascial fibers are immobilized, the structure of the fiber appears to loose crimp and springiness. Regular proper exercise loading of the fascial fibers may induce a more youthful collagen architecture with wavy fibers and increased elastic storage capacity (Schleip & Baker, 2015, p. 7).					

Fascia-focused Pilates training

Table 39.5 (Continued) The myofascial continuities^a and J. H. Pilates Reformer

Use this Pilates Reformer fascia-focused training guide to plan your Reformer classes and individual practice for thorough fascial training. Each exercise cultivates at least one of the fascia-focused movement criteria and activates one primary and several secondary myofascial continuities

Fascia-focused movement criteria	
18. Develop elastic recoil	
19. Stimulate tissue renewal	The renewal speed of the body-wide fascial network is quite slow, with a half-life between months and years rather than days or weeks. During the first 3 hours after appropriate exercise loading, collagen synthesis is increased. However, collagen degradation is also increased. During the first one and a half days following exercise loading collagen degradation outweighs collagen synthesis. Forty-eight hours after exercise, net synthesis of collagen production becomes positive (Schleip and Baker, 2015, pp. 8–9). The mobile, adaptable, fibrillar network with its intersecting fibers develops a mechanical harmony that is lost when healthy tissue is damaged. The body's repair mechanisms are unable to restore the fibrillar network in the damaged area to its original condition. The replacement tissue is of poor quality but can be improved by early mobilization of the injured area by manual therapy to enhance the flexibility of the scar tissue (Guinberteau and Armstrong, 2015, p. 162).
20. Develop tissue resilience	According to the bidensegrity model, the neuromyofascial system accommodates movement as a self-adjusting whole. The myofascial meridians map out a set of connective tissues that form the outer tensional network that pulls in on the skeleton. The "isolated muscle theory" that has predominated our thinking has limited our perception of this body-wide "give" that is essential to resilience (Myers, p. 46; Ch. 6 in Schleip & Baker, 2015).

^a *Anatomy Trains* by Thomas W. Myers (2020) introduces the term myofascial meridians. In this chapter I replace "meridians" with "continuities".

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Table 39.6 The myofascial continuities^a and fascia-focused Reformer exercises
Use this Pilates Reformer fascia-focused training guide to plan your Reformer classes and individual practice for thorough fascial training. Each exercise cultivates at least one of the fascia-focused movement criteria and activates one primary and several secondary myofascial continuities

Fascia-focused movement criteria		The myofascial continuities ^a							
Movement requirements		Biotensegrity model							
1. Movements are sustainable for motor control refinement and collagen remodeling	2. Movements encourage awareness of and embodiment of the biotensegrity model ^b	b "A tensegrity structure provides a global response to a local mechanical stress. The result is a degree of independence from the force of gravity. Without a tensegrity model, our fibrillar structure would collapse under gravitational force. With the tensegrity model, our structure absorbs and disperses the compression by spreading the load throughout the entire network, including the structures at the periphery" (Guimberteau and Armstrong, 2015). "The value of the Tensegrity model is not that it necessarily changes a particular approach to treatment but that it provides a better means to visualize the mechanics of the body in the light of new understandings about functional anatomy" (Scarr, 2014).							
		Superficial Back	Functional	Lateral	Spiral	Deep Front	Arm	Superficial Front	
3. Access all appropriate myofascial continuities in optimal sequence for client profile		Semicircles up & down with hip external and internal rotation	Hip joint mobility, supine leg abduction/adduction	Seated thoracic lateral flexion and rotation (Advanced Mermaid)	Hip joint mobility, standing lunge hip external/internal rotation	Quadruped hip flexion, emphasize Anterior Oblique Sling System	Seated thoracic lateral flexion and rotation (Advanced Mermaid)	Quadruped hip extension with lateral translation on the Rocker Reformer	
Movement intent									
4. Interoception — Interstitial nerves in fascia serve an interoceptive, rather than proprioceptive or nociceptive, function. Stimulation of those free nerve endings provides information about the condition of the body in search for homeostasis in relation to physiological needs. Interoceptive signaling is associated with feelings like warmth, nausea, hunger, soreness, effort, heaviness or lightness. Perceptions about internal somatic sensations are associated with emotional preferences and feelings (Schleip, 2015).									
5. Proprioceptive refinement — Proprioceptive sensations are connected with position, tendon and muscle sensations		Thoracic rotation with lateral translation, hands on shoulder rests, standing	Supine lateral translation (all three variations)	Quadruped lateral translation	Bridging with rotation	Long-stretch with lateral translation on Rocker Reformer	Thoracic rotation with lateral translation, hands on shoulder rests in quadruped and standing	Hip joint mobility, supine femur circles	

Fascia-focused Pilates training

Table 39.6 (Continued) The myofascial continuities* and fascia-focused Reformer exercises

Use this Pilates Reformer fascia-focused training guide to plan your Reformer classes and individual practice for thorough fascial training. Each exercise cultivates at least one of the fascia-focused movement criteria and activates one primary and several secondary myofascial continuities

Fascia-focused movement criteria							
6. Kinesthetic acuity — Kinesthesia (dynamic proprioception), the aptitude to sense the position and movement of the limbs and trunk	Quadruped hip extension with foot on foot bar, emphasize Anterior Oblique Sling System	Quadruped lateral translation	Thoracic rotation with lateral translation, hands on shoulder rests in quadruped and standing	Seated thoracic lateral flexion and rotation (Advanced Mermaid)	Elastic recoil arm jumps in quadruped	Elastic recoil arm jumps in quadruped	Hip joint mobility standing medial hip glide
Movement attributes							
7. Whole-body movements simultaneously involving large areas of the neuromyofascial system	Standing assisted squats standing on carriage on diagonal axis with respect to springs, thoracic and cervical rotation	Hip joint mobility, standing medial hip glide	The side kick kneeling position with foot on foot bar	Standing assisted squats standing on carriage on diagonal axis with respect to springs, thoracic rotation	Elastic recoil leg jumps, quadruped single leg jumps in hip extension	Semicircle Up & Down with rotation of pelvis and lumbar spine and hip external and internal rotation	Spine-massage with single cord to opposite foot
8. Whole-body continuity — Connect trunk to limbs and limbs to trunk — Connect deep structures to superficial ones and superficial structures to deep ones	Prone on long box, feet on foot bar, hip and knee extension with thoracic extension	Semicircles up & down with hip external and internal rotation	Standing side splits on a diagonal with respect to the springs, thoracic lateral flexion	Differentiated spine rotation standing on diagonal axis facing away from the foot bar holding a Magic Circle	Elastic recoil leg jumps, supine ankle jumps with knees extended	Standing assisted squats standing on carriage on diagonal axis with respect to springs, thoracic and cervical rotation	Quadruped lateral translation

(Continued)

Chapter 39

Table 39.6 (Continued) The myofascial continuities^a and fascia-focused Reformer exercises

Use this Pilates Reformer fascia-focused training guide to plan your Reformer classes and individual practice for thorough fascial training. Each exercise cultivates at least one of the fascia-focused movement criteria and activates one primary and several secondary myofascial continuities

Fascia-focused movement criteria								
9. Movement initiation — Connect proximal structures to distal ones and distal structures to proximal ones	Elastic recoil leg jumps, quadruped single leg jumps in hip extension	Supine lateral translation (all three variations)	Elastic recoil arm jumps seated single arm	Bridging with rotation	Hip joint mobility, supine femur circles	Elastic recoil arm jumps in quadruped	Elastic recoil leg jumps, quadruped lower leg jumps	
	Bridging with rotation	Long-stretch with lateral translation on Rocker Reformer	The side kick kneeling position with foot on foot bar, hip extension with thoracic rotation	Semicircles up & down with rotation of pelvis and lumbar spine	Semicircles up & down with hip external and internal rotation	Seated thoracic lateral flexion and rotation (Advanced Mermaid)	Spine-massage with single cord to opposite foot	
10. Flowing movement sequences								
11. Multidirectional movements with slight changes of angle	Elastic recoil leg jumps, quadruped single leg jumps in hip extension (all landing variations)	Quadruped hip extension with foot on foot bar (all variations)	Elastic recoil leg jumps, side lying single leg jumps (all landing variations)	The side kick kneeling position with foot on foot bar, hip extension with thoracic rotation	Elastic recoil leg jumps, supine ankle jumps with knees extended	Elastic recoil arm jumps in quadruped	Quadruped hip extension with lateral translation on the Rocker Reformer	
12. Force/load transfer — Movements that distribute force/load through the neuromyo-fascial system	Standing assisted squats, stand on floor on diagonal axis with respect to springs	The side kick kneeling position with foot on foot bar (all variations)	Elastic recoil arm jumps seated single arm	Thoracic rotation with lateral translation, hands on shoulder rests in quadruped and standing	Elastic recoil leg jumps, quadruped single leg jumps in hip extension	Differentiated spine rotation Standing on the diagonal axis facing toward the foot bar without Magic Circle	Spine-massage with single cord to opposite foot	

Fascia-focused Pilates training

Table 39.6 (Continued) The myofascial continuities* and fascia-focused Reformer exercises Use this Pilates Reformer fascia-focused training guide to plan your Reformer classes and individual practice for thorough fascial training. Each exercise cultivates at least one of the fascia-focused movement criteria and activates one primary and several secondary myofascial continuities							
Fascia-focused movement criteria		Elastic recoil arm jumps in quadruped	Standing assisted squats standing on carriage on diagonal axis with respect to springs, thoracic and cervical rotation	Thoracic rotation with lateral translation, hands on shoulder rests, standing	Standing assisted squats standing on carriage on diagonal axis with respect to springs, thoracic and cervical rotation	Thoracic rotation with lateral translation, hands on shoulder rests, standing	Standing assisted squats standing on carriage on diagonal axis with respect to springs, thoracic and cervical rotation
13. Preparatory counter movement		Elastic recoil arm jumps in quadruped	Standing assisted squats standing on carriage on diagonal axis with respect to springs, thoracic and cervical rotation	Thoracic rotation with lateral translation, hands on shoulder rests, standing	Standing assisted squats standing on carriage on diagonal axis with respect to springs, thoracic and cervical rotation	Thoracic rotation with lateral translation, hands on shoulder rests, standing	Standing assisted squats standing on carriage on diagonal axis with respect to springs, thoracic and cervical rotation
		Spine-massage with single cord to opposite foot	Elastic recoil leg jumps, quadruped single leg jumps in hip extension	Elastic recoil leg jumps, side lying single leg jumps	Elastic recoil arm jumps seated with trunk rotation	Elastic recoil arm jumps in quadruped	Spine-massage with single cord to opposite foot
14. Dynamic stretching — Slow and fast tempo variations		Standing side splits on a diagonal with respect to the springs	Standing side splits on a diagonal with respect to the springs, thoracic rotation	Standing side splits on a diagonal with respect to the springs	Standing side splits on a diagonal with respect to the springs	Standing side splits on a diagonal with respect to the springs	Spine-massage with single cord to opposite foot
		Elastic recoil leg jumps, quadruped lower leg jumps	Elastic recoil leg jumps, supine ankle jumps with knees extended	Elastic recoil leg jumps, supine ankle jumps with knees extended	Elastic recoil leg jumps, supine ankle jumps with knees extended	Elastic recoil leg jumps, supine ankle jumps with knees extended	Elastic recoil leg jumps, quadruped lower leg jumps
Movement outcomes		Approximately two-thirds of the volume of fascial tissues is made up of water. During application of mechanical load, whether by stretching or via local compression, a significant amount of water is pushed out of the more stressed zones. Application of external loading of fascial tissues can result in a refreshed hydration with improvement in the viscoelastic tissue properties (Schleip and Baker, 2015, p. 8).					
15. Facilitate tissue hydration		Approximately two-thirds of the volume of fascial tissues is made up of water. During application of mechanical load, whether by stretching or via local compression, a significant amount of water is pushed out of the more stressed zones. Application of external loading of fascial tissues can result in a refreshed hydration with improvement in the viscoelastic tissue properties (Schleip and Baker, 2015, p. 8).					

(Continued)

Chapter 39

Table 39.6 (Continued) The myofascial continuities^a and fascia-focused Reformer exercises
Use this Pilates Reformer fascia-focused training guide to plan your Reformer classes and individual practice for thorough fascial training. Each exercise cultivates at least one of the fascia-focused movement criteria and activates one primary and several secondary myofascial continuities

Fascia-focused movement criteria							
16. Develop glide within the three-dimensional neuromyofascial system	The normal glide of the collagen layers over one another may be hampered by loose connective tissue that has accumulated a variety of waste products. Failure of loose connective tissue gliding occurs in cases of overuse syndrome, trauma and surgery. Increase of local temperature contributes to fascial gliding. When the temperature is increased to over 40°C, the three-dimensional superstructure of hyaluronan chains break down progressively. This decreases hyaluronan viscosity of the loose connective tissue that is present within and beneath deep fasciae and muscles. The temporary increase in temperature can be provided by massage, movement or a warm shower. It is possible that during sleep, immobilization may cause increased hyaluronan viscosity and consequently a stiffness of the deep fasciae and muscles (Stecco, 2015, p. 60, p. 64).						
17. Develop regular lattice with crimp	The microstructure of collagen fibers shows undulations called crimp, which is reminiscent of elastic springs. When people age or when fascial fibers are immobilized, the structure of the fiber appears to loose crimp and springiness. Regular proper exercise loading of the fascial fibers may induce a more youthful collagen architecture with wavy fibers and increased elastic storage capacity (Schleip and Baker, 2015, p. 7).						
18. Develop elastic recoil	Elastic recoil leg jumps, quadruped lower leg jumps	Elastic recoil arm jumps in quadruped	The side kick kneeling position with foot on foot bar in hip flexion—jumping with foot on jump board	Elastic recoil arm jumps seated with trunk rotation	Elastic recoil leg jumps, supine ankle jumps with knees extended	Elastic recoil arm jumps in quadruped	Elastic recoil leg jumps, quadruped lower leg jumps
19. Stimulate tissue renewal	The renewal speed of the body-wide fascial network is quite slow, with a half-life between months and years rather than days or weeks. During the first three hours after appropriate exercise loading, collagen synthesis is increased. However, collagen degradation is also increased. During the first one and a half days following exercise loading, collagen degradation outweighs collagen synthesis. Forty-eight hours after exercise, net synthesis of collagen production becomes positive (Schleip and Baker, 2015, pp. 8–9). The mobile, adaptable, fibrillar network with its intersecting fibers develops a mechanical harmony that is lost when healthy tissue is damaged. The body's repair mechanisms are unable to restore the fibrillar network in the damaged area to its original condition. The replacement tissue is of poor quality but can be improved by early mobilization of the injured area by manual therapy to enhance the flexibility of the scar tissue (Guimberteau and Armstrong, 2015, p. 162).						
20. Develop tissue resilience	According to the biotensegrity model, the neuromyofascial system accommodates movement as a self-adjusting whole. The myofascial meridians map out a set of connective tissues that form the outer tensional network that pulls in on the skeleton. The "isolated muscle theory" that has predominated our thinking has limited our perception of this body-wide "give" that is essential to resilience (Myers, p 46; Ch. 6 in Schleip and Baker, 2015).						

^a *Anatomy Trains* by Thomas W. Myers (2020) introduces the term "myofascial meridians". In this chapter I replace "meridians" with "continuities".

Fascia-focused Pilates training

Fascial training inspires development of Pilates teaching techniques

Although the Pilates movement principles correlate with fascia-focused training criteria, the design of fascia training programs in the contemporary Pilates environment requires new consideration of all elements of Pilates program design. Research on the neuromyofascial system suggests new criteria for exercise sequencing, new choices of movement tempo and rhythm and different language for verbal cueing, as well as clarification of the quality and direction of touch for tactile cues.

The practice of training fascia in Pilates has been inspired by the publication of Thomas W. Myers' book (Myers, 2020) (Chapter 36). Pilates teachers seeking to augment their movement education with interdisciplinary study applied the 11 myofascial chains described by Myers (for an evidence-based review, see Chapter 36) to the Pilates mat and apparatus repertoire and began the paradigm shift from "isolated muscle theory" to the "longitudinal anatomy" (Myers, 2020). Figure 39.3 shows the 11 myofascial chains drawn on Pilates mat and apparatus exercises, illustrating the effectiveness of this movement system for fascia-focused training.

illustrating the effectiveness of this movement program for fascia-focused training. Table 39.7 provides the key to Figure 39.3, naming each myofascial chain according to the model detailed by Myers in *Anatomy Trains*.

Phillip Beach proposed the Contractile Field model of movement (Beach, 2010). "Seeing movement as whole organism fields of contractility that have evolved along functional pathways offers us fresh approaches to assessment and treatment of the moving body...Core patterns of contractility, allied to field-like behaviour, suggest new ways to understand human movement" (Beach, 2010). The Contractile Field theory has not yet exerted a significant influence on fascia-focused training in Pilates. This may be due, in part, to the complexity of his theories and the fact that *Muscles and Meridians* is less richly illustrated than *Anatomy Trains*. The publication in 2012 of *Fascia: The Tensional Network of The Human Body* (Schleip et al., 2012) inspired the author to pursue interdisciplinary study of the properties and function of the neuromyofascial system and apply this research to movement education in the contemporary Pilates environment.



Figure 39.3

Myofascial continuities drawn on Pilates mat and apparatus exercises illustrate the effectiveness of this movement system for fascia-focused training. Table 39.7 names each myofascial meridian, each exercise, and the equipment based on the original designs of J. H. Pilates.

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Table 39.7 Key to Figure 39.3. Each model in Figure 39.3 performs a Pilates exercise that activates several myofascial continuities. The primary myofascial continuity is identified here together with the exercise name and equipment evolved from the inventions of J. H. Pilates

Myofascial continuity* drawn on Pilates mat and apparatus exercise	Pilates exercise	Pilates equipment
1. The Deep Front myofascial continuity	Teaser reverse	Clara step barrel
2. The Superficial Back myofascial continuity	Long spine massage — advanced version	Reformer with long straps
3. The Superficial Front myofascial continuity	Thigh-Stretch with spine extension and left rotation	Reformer with tower push through bar
4. The Lateral myofascial continuity	Mermaid — advanced version	Reformer and short box
5. The Spiral myofascial continuity front section	Swan-Dive rotations	Ladder Barrel and dowel in hands
6. The Spiral myofascial continuity back section	Thigh-Stretch — advanced version with spine extension and right rotation	Reformer with short straps
7. The Front Functional myofascial continuity	Side lunge with weight shift — advanced version	Combo chair and rotator disc under right foot
8. The Arm myofascial continuity Left – Deep Front Arm myofascial continuity Right – Superficial Back Arm myofascial continuity	Kneeling arm work and thigh stretch — advanced version with spine extension and right rotation	Reformer with short straps
9. The Deep Front myofascial continuity	Bridging — advanced version	Trapeze table with push through bar
10. The Superficial Back myofascial continuity	Jumping on footplate in quadruped	Reformer with footplate (footplate is hidden behind model #6)
11. The Spiral myofascial continuity front section	Standing/side splits with diagonal orientation	Reformer with standing platform
12. The Back Functional myofascial continuity	Leg-Pull – Front	Mat

*Continuity is used here as a synonym for the term “myofascial meridian”, used by Myers (2020). In this chapter I replace “meridians” with “continuities”.

The Pilates movement principles of Precision, Control, Centering, Flow and Whole Body Movement all correlate with the fascia-focused training principle of Proprioceptive Refinement and Kinesthetic Acuity. Jaap van der Wal’s (2009) writing on proprioception (Schleip et al., 2012) may guide Pilates teachers who seek clarity on

the perception of movement. “To understand the mechanical and functional circumstances of the fascial role in connecting and in conveying stresses and in proprioception, it is therefore more important to know the architecture of the connective and muscle tissue than the regular anatomical order or topography” (Schleip et al., 2012).

Fascia-focused Pilates training

Table 39.8 Fascia-focused movement criteria and J. H. Pilates high back chair and Wunda chair examples

Fascia-focused movement criteria		
Movement requirements	Biotensegrity model	
1. Movements are sustainable for motor control refinement and collagen remodeling	* "A tensegrity structure provides a global response to a local mechanical stress. The result is a degree of independence from the force of gravity. Without a tensegrity model, our fibrillar structure would collapse under gravitational force. With the tensegrity model, our structure absorbs and disperses the compression by spreading the load throughout the entire network, including the structures at the periphery" (Guimberteau and Armstrong, 2015). "The value of the Tensegrity model is not that it necessarily changes a particular approach to treatment but that it provides a better means to visualize the mechanics of the body in the light of new understandings about functional anatomy" (Scarr, 2014).	
2. Movements encourage awareness of and embodiment of the biotensegrity model*		
	Pilates principles	J. H. Pilates high back chair and Wunda chair examples
3. Access all appropriate myofascial continuities in optimal sequence for client profile	Whole-body movement	Reverse Swan/Torso Press Sit (Supine atop Chair)
Movement intent		
4. Interoception — interstitial nerves in fascia serve an interoceptive, rather than proprioceptive or nociceptive function. Stimulation of those free nerve endings provides information about the condition of the body in search for homeostasis in relation to physiological needs. Interoceptive signaling is associated with feelings like warmth, nausea, hunger, soreness, effort, heaviness or lightness. Perceptions about internal somatic sensations are associated with emotional preferences and feelings (Schleip, 2015)	Breathing	Interoceptive sensations are unique to each individual
5. Proprioceptive refinement — proprioceptive sensations are connected with position, tendon and muscle sensations	Concentration Control	Seated Leg Pumps
6. Kinesthetic acuity — kinesthesia (dynamic proprioception), the aptitude to sense the position and movement of the limbs and trunk	Precision	Piano Lesson Back (Squat in external rotation back to Chair)
Movement attributes		
7. Whole-body movements simultaneously involving large areas of the neuromyofascial system	Whole-body movement Breathing	Side Arm Twist (Seated atop Chair)
8. Whole-body continuity — connect trunk to limbs and limbs to trunk while connecting deep structures to superficial ones and superficial structures to deep ones	Whole-body movement	Cat (Kneeling atop Chair)
9. Movement initiation — connect proximal structures to distal ones and distal structures to proximal ones	Centering Breathing	Pike/Teaser on Floor (Seated on floor in front of Chair)
10. Flowing movement sequences	Breathing	Tendon Stretch One Leg

(Continued)

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Table 39.8 (Continued) Fascia-focused movement criteria and J. H. Pilates high back chair and Wunda chair examples

Fascia-focused movement criteria		
11. Multidirectional movements with slight changes of angle	Balanced muscle development (optimal activation of the neuromyofascial system)	Jack-Knife from Floor and Corkscrew
12. Force/load transfer — movements that distribute force/load through the neuromyofascial system		Washer Woman Over the Chair (Standing behind Chair) Press Up with Handles Facing In
13. Preparatory counter movement	Rhythm	Swan Front/Chest Press (Prone atop Chair)
14. Dynamic stretching — slow and fast tempo variations	Rhythm	Seated Mermaid/Side Arm Sit
Movement outcomes		
15. Facilitate tissue hydration		
16. Develop glide within the three dimensional neuromyo-fascial system		
17. Develop regular lattice with crimp		
18. Develop elastic recoil	<i>J. H. Pilates high back chair and wunda chair exercises do not develop elastic recoil of the neuromyofascial system</i>	
19. Stimulate tissue renewal		
20. Develop tissue resilience		

Reframe Pilates concepts with new understanding of the structure and function of the neuromyofascial system

In fascia-oriented training one practices body movements that engage the longest possible myofascial chains (Müller and Schleip, 2011). This is in harmony with the Pilates principle of whole body movement. In our second week of development, the fascial network is a unified whole and remains a single, unifying and communicating network from birth until death (Earls and Myers, 2010). Fascia-focused training shaped by

research on the properties and function of fascia reinforces the Pilates practice of whole body movement. The “Powerhouse” of J. H. Pilates (Pilates, 1945) and the inner unit of core control (Lee, 2012) may be understood in the context of whole-body continuity by an understanding of the Deep Front Line myofascial chain (Myers, 2020) (Chapter 36).

The concept of lumbopelvic stability and the Pilates principle of balanced muscle development may be informed by biotensegrity, derived from “biology”, “tension” and “integrity” (Scarr, 2018). The internal balance of tension and compression supports structural integrity. Any deformation

Fascia-focused Pilates training

Table 39.9 Fascia-focused movement criteria and fascia-focused combo chair examples				
Fascia-focused movement criteria		Biotensegrity model		
Movement requirements				
1. Movements are sustainable for motor control refinement and collagen remodeling		* A tensegrity structure provides a global response to a local mechanical stress. The result is a degree of independence from the force of gravity. Without a tensegrity model, our fibrillar structure would collapse under gravitational force. With the tensegrity model, our structure absorbs and disperses the compression by spreading the load throughout the entire network, including the structures at the periphery" (Gumbertau and Armstrong, 2015).		
2. Movements encourage awareness of and embodiment of the biotensegrity model*		"The value of the Tensegrity model is not that it necessarily changes a particular approach to treatment but that it provides a better means to visualize the mechanics of the body in the light of new understandings about functional anatomy" (Scarr, 2014).		
		Pilates principles	J. H. Pilates high back chair and Wunda chair examples	Fascia-focused combo chair examples
3. Access all appropriate myofascial continuities in optimal sequence for client profile		Whole-body movement	Reverse Swan/Torso Press Sit (Supine atop Chair)	Stand on rotator discs, one hand per pedal, thoracic and cervical rotation
Movement intent		Breathing	Interoceptive sensations are unique to each individual	
4. Interoception — Interstitial nerves in fascia serve an interoceptive, rather than proprioceptive or nociceptive function. Stimulation of those free nerve endings provides information about the condition of the body in search for homeostasis in relation to physiological needs. Interoceptive signaling is associated with feelings like warmth, nausea, hunger, soreness, effort, heaviness or lightness. Perceptions about internal somatic sensations are associated with emotional preferences and feelings (Schleip, 2015)				
5. Proprioceptive refinement — proprioceptive sensations are connected with position, tendon and muscle sensations		Concentration Control	Seated Leg Pumps	Standing ankle plantar/dorsi flexion hip neutral, external/internal rotation
6. Kinesthetic acuity — kinesesthesia (dynamic proprioception), the aptitude to sense the position and movement of the limbs and trunk		Precision	Piano Lesson Back (Squat in external rotation back to Chair)	Side to chair, one foot per pedal, inside foot in front, hands on top of chair in seven different positions
Movement attributes				
7. Whole-body movements simultaneously involving large areas of the neuromyofascial system		Whole-body movement Breathing	Side Arm Twist (Seated atop Chair)	Standing pelvic unleveling with ankle plantar flexion and thoracic lateral flexion and rotation

(Continued)

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Table 39.9 (Continued) Fascia-focused movement criteria and fascia-focused combo chair examples

Fascia-focused movement criteria			
8. Whole-body continuity — connect trunk to limbs and limbs to trunk while connecting deep structures to superficial ones and superficial structures to deep ones	Whole-body movement	Cat (Kneeling atop Chair)	Side to chair, one foot per pedal, outside foot in front, hands on top of chair in seven different positions
9. Movement initiation — connect proximal structures to distal ones and distal structures to proximal ones	Centering Breathing	Pike/Teaser on Floor (Seated on floor in front of Chair)	Seated spine extension with rotation
10. Flowing movement sequences	Breathing	Tendon Stretch One Leg	Seated lateral spine flexion with rotation
11. Multidirectional movements with slight changes of angle	Balanced muscle development (optimal activation of the neuromyo-fascial system)	Jack-Knife from Floor and Corkscrew	Facing chair, one foot per pedal, hands on wobble board, variation one—turn towards leg with knee flexion, variation two—turn away from leg with knee flexion
12. Force/load transfer — movements that distribute force/load through the neuromyo-fascial system		Washer Woman Over the Chair (Standing behind Chair) Press Up with Handles Facing In	One hand on rotator disc, one hand on chair pedal, single leg plank Facing chair, one foot per pedal, hands on top of chair in seven different positions
13. Preparatory counter movement	Rhythm	Swan Front/Chest Press (Prone atop Chair)	Seated lateral spine flexion with rotation (side bend away from pedal before side bending toward the pedal)
14. Dynamic stretching — slow and fast tempo variations	Rhythm	Seated Mermaid/Side Arm Sit	Seated spine extension with rotation
Movement outcomes			
15. Facilitate tissue hydration			
16. Develop glide within the three dimensional neuromyo-fascial system			
17. Develop regular lattice with crimp			
18. Develop elastic recoil		The high back chair and Wunda chair do not support elastic recoil training	The combo chair does not support elastic recoil training
19. Stimulate tissue renewal			
20. Develop tissue resilience			

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Table 39.10 Fascia-focused movement criteria and J. H. Pilates trapeze table examples		
Fascia-focused movement criteria		Biotensegrity model
Movement requirements		
1. Movements are sustainable for motor control refinement and collagen remodeling		* "A tensegrity structure provides a global response to a local mechanical stress. The result is a degree of independence from the force of gravity. Without a tensegrity model, our fibrillar structure would collapse under gravitational force. With the tensegrity model, our structure absorbs and disperses the compression by spreading the load throughout the entire network, including the structures at the periphery" (Gumbertau and Armstrong, 2015). "The value of the Tensegrity model is not that it necessarily changes a particular approach to treatment but that it provides a better means to visualize the mechanics of the body in the light of new understandings about functional anatomy" (Scarr, 2014).
2. Movements encourage awareness of and embodiment of the biotensegrity model*		
Pilates principles		J. H. Pilates trapeze table examples
3. Access all appropriate myofascial continuities in optimal sequence for client profile	Whole-body movement	Rolling In and Out
Movement intent		
4. Interoception — Intersitil nerves in fascia serve an interoceptive, rather than proprioceptive or nociceptive function. Stimulation of those free nerve endings provides information about the condition of the body in search for homeostasis in relation to physiological needs. Interoceptive signaling is associated with feelings like warmth, nausea, hunger, soreness, effort, heaviness or lightness. Perceptions about internal somatic sensations are associated with emotional preferences and feelings (Schleip, 2015)	Breathing	Interoceptive sensations are unique to each individual
5. Proprioceptive refinement — proprioceptive sensations are connected with position, tendon and muscle sensations	Concentration Control	Thigh Stretch
6. Kinesthetic acuity — kinesthesia (dynamic proprioception), the aptitude to sense the position and movement of the limbs and trunk	Precision	Magician
Movement attributes		
7. Whole-body movements simultaneously involving large areas of the neuromyofascial system	Whole-body movement Breathing	Hanging Down
8. Whole-body continuity — connect trunk to limbs and limbs to trunk while connecting deep structures to superficial ones and superficial structures to deep ones	Whole-body movement	Parakeet

(Continued)

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Table 39.10 (Continued) Fascia-focused movement criteria and J. H. Pilates trapeze table examples

Fascia-focused movement criteria			
9. Movement initiation — connect proximal structures to distal ones and distal structures to proximal ones	Centering Breathing		Push-Through Seated Back
10. Flowing movement sequences	Breathing		Spread Eagle
11. Multidirectional movements with slight changes of angle	Balanced muscle development (optimal activation of the neuromyofascial system)		Butterfly
12. Force/load transfer — movements that distribute force/load through the neuromyofascial system			Tower Hanging Up
13. Preparatory counter movement	Rhythm		Side-Lying Bicycle
14. Dynamic stretching — slow and fast tempo variations	Rhythm		Mermaid
Movement outcomes			
15. Facilitate tissue hydration			
16. Develop glide within the three dimensional neuromyofascial system			
17. Develop regular lattice with crimp			
18. Develop elastic recoil	J. H. Pilates trapeze table exercises do not develop elastic recoil of the neuromyofascial system		
19. Stimulate tissue renewal			
20. Develop tissue resilience			

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Table 39.11 Fascia-focused movement criteria and fascia-focused trapeze table examples

Fascia-focused movement criteria		Biotensegrity model		
Movement requirements				
1. Movements are sustainable for motor control refinement and collagen remodeling		*A tensegrity structure provides a global response to a local mechanical stress. The result is a degree of independence from the force of gravity. Without a tensegrity model, our fibrillar structure would collapse under gravitational force. With the tensegrity model, our structure absorbs and disperses the compression by spreading the load throughout the entire network, including the structures at the periphery” (Guimberteau and Armstrong, 2015). “The value of the Tensegrity model is not that it necessarily changes a particular approach to treatment but that it provides a better means to visualize the mechanics of the body in the light of new understandings about functional anatomy” (Scarr, 2014).		
2. Movements encourage awareness of and embodiment of the biotensegrity model*				
		Pilates principles	J. H. Pilates trapeze table examples	Fascia-focused trapeze table examples
3. Access all appropriate myofascial continuities in optimal sequence for client profile		Whole-body movement	Rolling In and Out	Prone scapula patterning
4. Movement intent				
5. Interception — Interstitial nerves in fascia serve an interoceptive, rather than proprioceptive or nociceptive function. Stimulation of those free nerve endings provides information about the condition of the body in search for homeostasis in relation to physiological needs. Interoceptive signaling is associated with feelings like warmth, nausea, hunger, soreness, effort, heaviness or lightness. Perceptions about internal somatic sensations are associated with emotional preferences and feelings (Schleip, 2015)		Breathing	Interoceptive sensations are unique to each individual	
6. Proprioceptive refinement — proprioceptive sensations are connected with position, tendon and muscle sensations		Concentration Control	Thigh Stretch	Seated on rotator disc scapular depression, elbow flexion, arm external rotation
7. Kinesthetic acuity — kinesthesia (dynamic proprioception), the aptitude to sense the position and movement of the limbs and trunk		Precision	Magician	Supine to side lying assisted trunk rotation with arm internal and external rotation
8. Movement attributes				
9. Whole-body movements simultaneously involving large areas of the neuromyofascial system		Whole-body movement Breathing	Hanging Down	Seated “circle-saw”

(Continued)

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Table 39.11 (Continued) Fascia-focused movement criteria and fascia-focused trapeze table examples

Fascia-focused movement criteria			
10. Whole-body continuity — connect trunk to limbs and limbs to trunk while connecting deep structures to superficial ones and superficial structures to deep ones	Whole-body movement	Parakeet	Parakeet with single hip flexion
11. Movement initiation — connect proximal structures to distal ones and distal structures to proximal ones	Centering Breathing	Push-Through Seated Back	Supine assisted spine extension with arms crossed in external rotation
12. Flowing movement sequences	Breathing	Spread Eagle	Bridge with knees extended
13. Multidirectional movements with slight changes of angle	Balanced muscle development (optimal activation of the neuromyofascial system)	Butterfly	Seated on rotator disc scapular depression, elbow flexion, arm external rotation
14. Force/load transfer — movements that distribute force/load through the neuromyofascial system		Tower Hanging Up	Prone scapula patterning Parakeet with double knee flexion and extension
15. Preparatory counter movement	Rhythm	Side-Lying Bicycle	Supine to side lying assisted trunk rotation with arm internal and external rotation
16. Dynamic stretching — slow and fast tempo variations	Rhythm	Mermaid	Seated "circle-saw"
17. Movement outcomes			
18. Facilitate tissue hydration			
19. Develop glide within the three dimensional neuromyofascial system			
20. Develop regular lattice with crimp			
21. Develop elastic recoil		The trapeze table does not support elastic recoil training.	
22. Stimulate tissue renewal			
23. Develop tissue resilience			

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Table 39.12 Fascia-focused movement criteria and J. H. Pilates spine-corrector examples			
Fascia-focused movement criteria		Biotensegrity model	
Movement requirements			
1. Movements are sustainable for motor control refinement and collagen remodeling		* "A tensegrity structure provides a global response to a local mechanical stress. The result is a degree of independence from the force of gravity. Without a tensegrity model, our fibrillar structure would collapse under gravitational force. With the tensegrity model, our structure absorbs and disperses the compression by spreading the load throughout the entire network, including the structures at the periphery" (Guimberteau and Armstrong, 2015). "The value of the Tensegrity model is not that it necessarily changes a particular approach to treatment but that it provides a better means to visualize the mechanics of the body in the light of new understandings about functional anatomy" (Scarr, 2014).	
2. Movements encourage awareness of and embodiment of the biotensegrity model*			
3. Access all appropriate myofascial continuities in optimal sequence for client profile		Pilates principles	J. H. Pilates spine-corrector examples
Movement intent		Whole-body movement	Back Arch and Bridge
4. Interception — Interstitial nerves in fascia serve an interoceptive, rather than proprioceptive or nociceptive function. Stimulation of those free nerve endings provides information about the condition of the body in search for homeostasis in relation to physiological needs. Interoceptive signaling is associated with feelings like warmth, nausea, hunger, soreness, effort, heaviness or lightness. Perceptions about internal somatic sensations are associated with emotional preferences and feelings (Schleip, 2015).		Breathing	Interoceptive sensations are unique to each individual
5. Proprioceptive refinement — proprioceptive sensations are connected with position, tendon and muscle sensations		Concentration Control	Balance
6. Kinesthetic acuity — kinesthesia (dynamic proprioception), the aptitude to sense the position and movement of the limbs and trunk		Precision	Swimming
Movement attributes			
7. Whole-body movements simultaneously involving large areas of the neuromyofascial system		Whole-body movement Breathing	Teaser
8. Whole-body continuity — connect trunk to limbs and limbs to trunk while connecting deep structures to superficial ones and superficial structures to deep ones		Whole-body movement	Helicopter

(Continued)

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Table 39.12 (Continued) Fascia-focused movement criteria and J. H. Pilates spine-corrector examples

Fascia-focused movement criteria		
9. Movement initiation — connect proximal structures to distal ones and distal structures to proximal ones	Centering Breathing	Swan
10. Flowing movement sequences	Breathing	Rolling In and Out
11. Multidirectional movements with slight changes of angle	Balanced muscle development (optimal activation of the neuromyofascial system)	Corkscrew
12. Force/load transfer — movements that distribute force/load through the neuromyofascial system		The Rollover with Legs Spread (Both Ways) Grasshopper
13. Preparatory counter movement	Rhythm	Low Bridge
14. Dynamic stretching — slow and fast tempo variations	Rhythm	Scissors
Movement outcomes		
15. Facilitate tissue hydration		
16. Develop glide within the three dimensional neuromyofascial system		
17. Develop regular lattice with crimp		
18. Develop elastic recoil	J. H. Pilates spine-corrector exercises do not develop elastic recoil of the neuromyofascial system	
19. Stimulate tissue renewal		
20. Develop tissue resilience		

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Table 39.13 Fascia-focused movement criteria and J. H. Pilates ladder barrel examples			
Fascia-focused movement criteria		Biotensegrity model	
Movement requirements			
1. Movements are sustainable for motor control refinement and collagen remodeling		* "A tensegrity structure provides a global response to a local mechanical stress. The result is a degree of independence from the force of gravity. Without a tensegrity model, our fibrillar structure would collapse under gravitational force. With the tensegrity model, our structure absorbs and disperses the compression by spreading the load throughout the entire network, including the structures at the periphery" (Guimberteau and Armstrong, 2015).	
2. Movements encourage awareness of and embodiment of the biotensegrity model*		"The value of the Tensegrity model is not that it necessarily changes a particular approach to treatment but that it provides a better means to visualize the mechanics of the body in the light of new understandings about functional anatomy" (Scarr, 2014).	
Pilates principles		J. H. Pilates ladder barrel examples	
3. Access all appropriate myofascial continuities in optimal sequence for client profile		Whole-body movement	Horseback
Movement intent			
4. Interception — Intersitital nerves in fascia serve an interoceptive, rather than proprioceptive or nociceptive function. Stimulation of those free nerve endings provides information about the condition of the body in search for homeostasis in relation to physiological needs. Interoceptive signaling is associated with feelings like warmth, nausea, hunger, soreness, effort, heaviness or lightness. Perceptions about internal somatic sensations are associated with emotional preferences and feelings (Schleip, 2015).		Breathing	Interoceptive sensations are unique to each individual
5. Proprioceptive refinement — proprioceptive sensations are connected with position, tendon and muscle sensations		Concentration Control	Side Sit-Ups
6. Kinesthetic acuity — kinesthesia (dynamic proprioception), the aptitude to sense the position and movement of the limbs and trunk		Precision	Twist
Movement attributes			
7. Whole-body movements simultaneously involving large areas of the neuromyofascial system		Whole-body movement Breathing	Swan Dive
8. Whole-body continuity — connect trunk to limbs and limbs to trunk while connecting deep structures to superficial ones and superficial structures to deep ones		Whole-body movement	Grasshopper

(Continued)

Table 39.13 (Continued) Fascia-focused movement criteria and J. H. Pilates ladder barrel examples

Fascia-focused movement criteria		
9. Movement initiation — connect proximal structures to distal ones and distal structures to proximal ones	Centering Breathing	Bicycle
10. Flowing movement sequences	Breathing	Climb-a-Tree
11. Multidirectional movements with slight changes of angle	Balanced muscle development (optimal activation of the neuromyofascial system)	Helicopter
12. Force/load transfer — movements that distribute force/load through the neuromyofascial system		Horseback Stomach Jumps
13. Preparatory counter movement	Rhythm	Scissors
14. Dynamic stretching — slow and fast tempo variations	Rhythm	Back Bend to Forward Bend
Movement outcomes		
15. Facilitate tissue hydration		
16. Develop glide within the three dimensional neuromyofascial system		
17. Develop regular lattice with crimp		
18. Develop elastic recoil	J. H. Pilates barrel exercises do not develop elastic recoil of the neuromyofascial system	
19. Stimulate tissue renewal		
20. Develop tissue resilience		

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will create strain that is distributed throughout the body. Any injury rapidly becomes a strain distribution patterned into the whole body that requires a whole-body assessment and whole-body treatment. Pilates teachers are challenged to see the whole body in terms of biotensegrity, in order to develop coherent movement program designs that address the neuromyofascial system and develop motor control in support of functional movement.

Fascia-focused training has the criteria of Proximal Initiation. This correlates with the Pilates principle of Centering. J. H. Pilates used the term “Powerhouse” to refer to the girdle of strength between the top of the pelvis and the bottom of the ribcage responsible for Centering or Proximal Initiation. Contemporary Pilates teacher education, influenced by Diane Lee (2011), explains that the inner unit of core control is the key to efficient, graceful, balanced movement (St. John, 2013).

The thoracolumbar fascia is essential for lumbar spine stability and force closure of the sacroiliac joint

The thoracolumbar fascia (TLF) is a three-dimensional girdling structure consisting of several aponeurotic and fascial layers that separates the paraspinal muscles from the muscles of the posterior abdominal wall. Several muscles attach to the TLF and its thick composite, which attaches to the posterior superior iliac spine and the sacrotuberous ligament. The thoracolumbar composite assists in maintaining the integrity of the lower lumbar spine and the sacroiliac joint. The latissimus dorsi, gluteus maximus and abdominal muscles, primarily the transversus

abdominis, attach to the TLF. Tension applied by these muscles, especially the transversus abdominis, can be transmitted through the TLF to beneficially stiffen the lumbar spine, creating stability, and increase the force closure of the sacroiliac joint. Flexion of the spine stretches the TLF, diminishing its lateral dimensions. Resistance to lateral retraction of the TLF by the abdominal muscles will stiffen this tissue and increase resistance to flexion of the lumbar region. Contraction of the paraspinal muscles increases intracompartmental pressure and contributes to the hydraulic amplifier effect supporting the lumbar spine. Increased tone in the lumbar multifidus muscle should increase the tension created by the thoracolumbar composite between the two posterior superior iliac spines. This increased, medially directed tension leads to force closure of the sacroiliac joint, thus stabilizing the pelvis (Willard et al., 2012).

As Pilates teachers advance their interdisciplinary education, aligning Pilates movement principles and practices with recent findings in motor learning, neuroscience and fascia-focused movement, it is likely that the movement form founded by J. H. Pilates will continue to attract practitioners worldwide (Larkam, 2017).

Augment fascia-focused Pilates training with fascial release for structural balance

Fascia-focused Pilates training may be integrated with manual therapy or fascial release for structural balance. When one practitioner has acquired both skill sets, the client will be fortunate enough to receive fascia-focused training for motor control and movement education together

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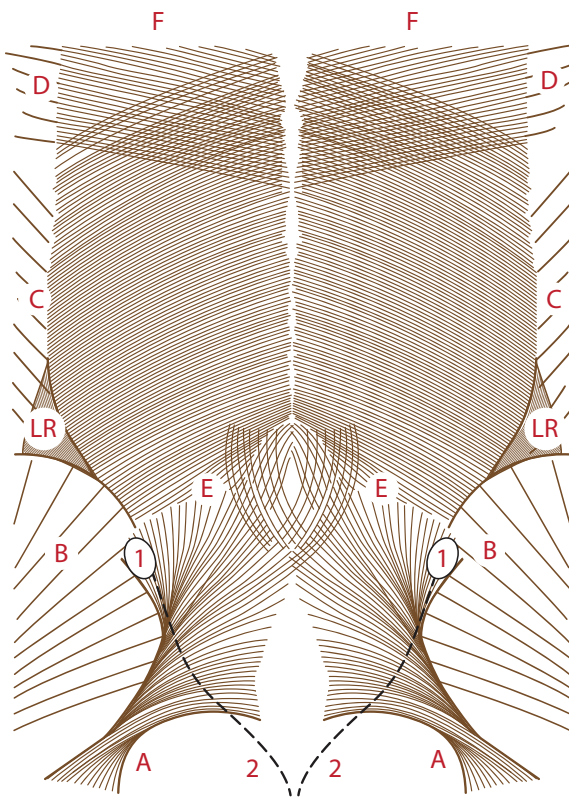


Figure 39.4

The deep layer of the thoracolumbar fascia and different fiber directions of attachments to: (A) sacrotuberous ligament connecting to hamstrings, (B) fascia of gluteus medius, (C) fascia of internal oblique, (D) serratus posterior inferior, (E) erector spinae muscles, (1) posterior superior iliac spine, (2) sacrum, (LR) lateral raphe.

Reproduced with permission from Schleip, R. & Baker, A. (Eds) (2015). *Fascia in Sport and Movement*. © Handspring Publishing.

with fascia release for structural balancing. If the practitioner has only one skill set, the client may seek two practitioners who can provide complementary sessions. Precise, whole body movement is necessary for training the fascia

but may not be sufficient to facilitate efficient movement. The body as a biotensegrity system responds to trauma, contracting and retracting around all axes (Earls and Myers, 2010). When fascia release for structural balance opens the body in one dimension, it seems to respond in all dimensions.

Design tactile cues to reinforce fascia-focused Pilates training

The tactile cues modeled by Joseph Pilates in his films (Pilates, 1932–1945) indicate a strong touch, sometimes forcefully pushing the client into the intended position. Fascia-focused training encourages a perceptual refinement of shear, gliding and tensioning motions in superficial fascial membranes (Müller & Schleip, 2011) (Chapter 24). This is inspired by the finding that the superficial fascia layers are more densely populated with mechanoreceptive nerve endings than the tissue situated more internally (Stecco et al., 2014) (Chapter 15). Although this perceptual refinement is recommended in terms of movement, it seems reasonable to apply this discovery to shaping tactile cues for fascia-focused Pilates training. By applying precise vectors at the depth of the superficial fascia layer, conveying myofascial continuity to a bony landmark or organ, a clear direction in space can be given. For example, in Figure 39.3, in Exercise no. 12, the desired thoracic kyphosis has been lost. Accurate spinal organization may be reinforced by placing the pads of the four fingers of the teacher's hand on the posterior spinous processes of the thoracic vertebrae between the scapulae. Instruct the client to increase pressure from the little finger side of the heel of the hand into the mat as she brings the sternum up, in the direction of the

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vertebrae, bringing the spinous processes convex toward the ceiling.

Another example of a tactile cue harmonious with fascia-focused training involves exercise no. 10 in Figure 39.3. Support of the thoracolumbar junction has been lost, resulting in compression in the area of T11, T12 and L1. The palm surface of the instructor's fingers are placed on the anterior lateral surfaces of the lower ribs, guiding the ribs towards the back of the thorax, asking the client to draw the lower ribs in and up toward the ceiling. Tactile cues are to be used judiciously, to inform or encourage rather than to force or overwhelm. In fascia-focused Pilates training, the client is the active agent who has the responsibility of shaping the movement rather than the role of a passive recipient of instructor actions.

Fascia-focused Pilates training is a young field undergoing rapid development

The intentional practice of fascia-focused Pilates training has only been in development since 2001. It has not been scientifically evaluated. The principles of fascia-focused training may inform all elements of Pilates program design, including utilization of all planes of motion, movement sequencing, tempo, duration, frequency, selection of resistance, verbal cues and tactile cues.

The work of the international community of Pilates teachers, who are evolving the field of fascia-focused Pilates training, occurs in Pilates studios throughout the world. In these movement laboratories, Pilates teachers engage in interdisciplinary enquiry and collaborate with clients to facilitate functional, elegant movement in activities of daily living, athletics and performance.

Practical application

Transform a J. H. Pilates mat exercise, that addresses muscular function, into a Pilates reformer sequence that stimulates fascial tissues, satisfying the principles of fascia-focused training. In Figure 39.3, exercise no. 12, a Leg Pull Front is demonstrated. This J. H. Pilates mat exercise requires integration of all the Myofascial chains (Myers, 2020) (Chapter 36). However, Leg Pull Front does not satisfy the Fascia-focused training criteria (Table 39.2) (Chapter 24). This demanding mat exercise provides limited opportunities for Preparatory Counter-Movement, Flowing Movement Sequences, Tempo Variation and Multidirectional Movements, with slight changes in angle that lead to Tissue Hydration and Renewal. Leg Pull Front does not involve elastic recoil (Chapter 8). Contrast Leg Pull Front to the Fascia-focused Pilates Reformer exercise shown in Figure 39.3, exercise no. 10. Jumping on the Footplate in Quadruped requires integrated action of all the Myofascial chains utilized in Leg Pull Front. (The Footplate, or Jump Board, is hidden by the person doing Exercise 6.) One blue spring connects the reformer carriage to the frame. In addition to fulfilling the myofascial chain requirements of Fascia-focused training, exercise no. 10 satisfies a number of Fascia-focused criteria. The dancer's demi plié or knee bend prepares for the ankle plantar flexion and knee extension required for each jump. This is Preparatory Counter-Movement (Chapter 24). Landing from each jump involves deceleration through the entire lower kinematic chain together with enhanced tensioning of the long fascial chains that connect the feet with the pelvic girdle, spine, thorax, shoulder girdle, arms and hands. This creates a flowing movement sequence. The jumps may be quick and small or slow and large. This is Tempo Variation. The jumping leg may be oriented in neutral hip rotation, external rotation or internal rotation. This fulfills Multidirectional Movements with slight changes in angle. The elastic recoil property of Reformer Jumping in quadruped makes additional demands on proprioceptive refinement and kinesthetic acuity, resulting in tissue hydration and tissue renewal.

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Further reading

Note: Elizabeth Larkam has produced a compilation of short videos to help you gain a deeper understanding. Watch these videos free of charge at www.pilatesanytime.com/fascia. To redeem your free access use code FASCIA

These short videos were compiled as a companion to her book *Fascia in Motion: Fascia-focused Movement for Pilates* (Handspring, 2017).

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Blom, M.-J. (2012) Pilates and Fascia: The art of 'working in'. *Fascia: The Tensional Network of The Human Body*. 7 (22): 451–456. Churchill Livingstone Elsevier.

These five contemporary Pilates classes reflect the influence of Fascia-focused training and the myofascial continuities discussed in the chapter. Viewers have a 30-day free trial on the Pilates Anytime website with the Code: LARKAM.

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